Supported Two-Phase Reading (STR) in L2 Acquisition: Theoretical Foundations, Protocol Architecture, and Research Agenda

Abstract

Supported Two-Phase Reading (STR) is proposed as a cognitively optimized framework for second language (L2) reading that explicitly couples secure initial comprehension with mandatory retrieval of meaning. In Phase I (Supported Reading), learners process a passage with full or near-full in-flow support (interlinear translations, brief glosses, and optional multimodal cues), minimizing extraneous cognitive load and stabilizing a coherent situation model of the text. In Phase II (Consolidation Practice), all aids are withdrawn categorically or systematically faded, and the same passage is reread without support, requiring learners to reconstruct meaning from memory and thereby operationalizing retrieval practice in a naturalistic reading context.

The article situates STR at the intersection of Cognitive Load Theory, Vygotskian sociocultural theory and the Zone of Proximal Development, dual-channel and multimedia learning theories, and ecological / neuro-interactive perspectives that treat reading as action rather than passive intake. It formalizes STR as a protocolized sequence with clearly defined constructs such as the L1-Dependency Index, Autonomy Score, and Fading Threshold, and outlines how these can be measured in future research. In addition, the paper sketches an AI-enabled implementation pathway in which large language models and neuro-adaptive indicators (e.g., pupillometry) support adaptive scaffolding and data-rich evaluation. As a conceptual and positioning contribution, the article advances STR as a next-generation, empirically testable model for integrating comprehension assurance and retrieval-based consolidation in L2 reading instruction.

Originality and Contribution

Status of the Work
This article is conceptual and positioning in nature. It does not report original empirical data but develops Supported Two-Phase Reading (STR) as a theoretically grounded, protocolized framework for L2 reading. To the best of current knowledge, it is the first academic operationalization of a two-phase reading process that combines: (a) a fully supported Phase I with dense in-flow assistance, (b) a pre-specified, categorical fading event from 100% to 0% support, (c) a mandatory second pass over the same passage without any aids, and (d) process-oriented measurement of L1-dependency and emerging autonomy. Claims about effectiveness are formulated as testable hypotheses and are intended to guide preregistered empirical studies rather than to assert established effects.

Theoretical Contribution
The paper consolidates and extends the theoretical foundations of STR by:

- bridging input-oriented and memory-oriented perspectives, explicitly linking comprehension assurance in Phase I with retrieval-based consolidation in Phase II;
- integrating CLT, ZPD-based scaffolding, dual-channel / multimedia learning, and ecological views of reading into a single two-phase architecture;
- reframing Comprehensible Input as a necessary *precondition* for, rather than the primary driver of, long-term learning, thereby aligning STR with contemporary neurocognitive evidence on the role of retrieval.

Methodological Contribution
Methodologically, the article:

- defines STR as a standardized, passage-bound protocol (P1 → fading → P2) suitable for controlled comparisons under time- and exposure-matched conditions;
- introduces measurable constructs such as the L1-Dependency Index, Autonomy Score, and Fading Threshold as anchors for longitudinal and multi-method validation;
- outlines adherence and manipulation checks for the fading event (e.g., layout audits, log data, optional eye-tracking), and sketches how AI-based STR platforms can embed these checks by design, enabling preregistration, replication, and meta-analytic synthesis.

Significance
By combining a clearly specified protocol structure, a rich theoretical grounding, and an explicit research agenda, the article provides a coherent starting point for a sustained program of empirical work on two-phase reading. At the same time, its emphasis on AI-supported scaffolding and neuro-adaptive measurement highlights a practical pathway for transforming STR from a conceptual model into a scalable, evidence-based component of L2 reading pedagogy.

1. Introduction

Second language reading places unusually high demands on the learner's cognitive system. Whereas first language competence typically emerges through largely implicit pattern learning, second language acquisition (SLA) requires the concurrent management of decoding, syntactic parsing, lexical inference, and meaning construction within a severely capacity-limited working memory. Existing pedagogical traditions tend to align with one of two poles: either they emphasize comprehensible input and scaffolding to secure immediate understanding, or they prioritize retrieval-based activities designed to promote long-term retention. These strands are rarely orchestrated within a single, explicitly structured architecture. As a result, reading instruction is often fragmented, learning outcomes are inconsistent, and cognitive resources are allocated suboptimally.

The Supported Two-Phase Reading (STR) model is introduced as a theoretically grounded, protocolized, and empirically testable response to this problem. Rather than proposing yet another isolated technique, STR formalizes a two-phase reading cycle that couples secure comprehension with obligatory retrieval. It systematically integrates Cognitive Load Theory (CLT), Vygotskian sociocultural theory and the Zone of Proximal Development (ZPD), dual-channel theories of multimedia learning, and the retrieval practice literature. Its central contribution lies not in inventing glossing, textual enhancement, or rereading, but in specifying **how** these elements are combined, sequenced, and measured: as a two-phase protocol with clearly defined constructs, transition criteria, and built-in options for adaptive and AI-supported personalization.

In Phase I (Supported Reading), learners work with the target text under full or near-full in-flow support. Interlinear glosses, L1 mediation, multimodal cues, and targeted enhancement are used to reduce extraneous cognitive load and guarantee accurate initial comprehension, even when texts slightly exceed the learner's unaided level. In Phase II (Consolidation Practice), these aids are withdrawn categorically or systematically faded, creating a condition of desirable difficulty in which meaning must be reconstructed from memory. This enforced, aid-free rereading operationalizes retrieval practice, supports the development of stable lexical and constructional representations, and fosters emerging reading autonomy. Pedagogically, the innovation of STR lies in treating the **phase transition**—the moment at which support is removed—as a deliberate design variable rather than an incidental side-effect of practice.

Beyond combining established theories, STR also draws on contemporary cognitive neuroscience and ecological perspectives on learning. It rejects a view of reading as passive intake and instead conceptualizes it as a dynamic interaction between learner, text, and environment. Phase I structures the

environment to afford comprehension; Phase II restructures it to afford autonomous retrieval and meaning reconstruction. In doing so, STR aligns with modern accounts that portray learning as action-driven, embodied, and contextually situated, rather than as the mere accumulation of decoded input.
The present article consolidates and extends earlier conceptual work on STR along three main axes. First, it refines the theoretical foundations of the model, situating it in relation to legacy frameworks such as Krashen's Comprehensible Input Hypothesis and incorporating critiques from contemporary SLA and neurocognitive research. Second, it specifies the STR protocol architecture in operational terms, introducing constructs such as the L1-Dependency Index and the Fading Threshold as targets for empirical measurement. Third, it outlines a research agenda and technological infrastructure—particularly AI-driven scaffolding and neuro-adaptive measurement tools—needed to move STR from a promising concept to a scalable, evidence-based pedagogical system.
By anchoring STR in established theory, integrating recent empirical insights, and articulating a clear path from protocol to implementation and validation, the article positions Supported Two-Phase Reading as a next-generation, cognitively optimized model for second language reading instruction.
Third, it outlines a research agenda comprising **five falsifiable hypotheses** targeting L1-dependency trajectories, retention profiles, cognitive load redistribution, and motivational outcomes.

2. Theoretical Foundations and Conceptual Rationale

The Supported Two-Phase Reading (STR) model draws its explanatory strength from the convergence of several major traditions in cognitive science and second language acquisition (SLA). These traditions—often discussed in parallel rather than in concert—include Cognitive Load Theory, sociocultural and ZPD-based views of learning, input-oriented SLA models, dual-channel and multimedia learning theories, and more recent ecological and neuro-interactive perspectives. STR integrates these strands into a single instructional architecture that explicitly manages cognitive load, structures scaffolding, and targets memory consolidation.
This section outlines the theoretical foundations that justify STR's two-phase design. It first explains how Cognitive Load Theory and sociocultural scaffolding jointly motivate the distinction between a supported comprehension phase and a retrieval-oriented consolidation phase. It then revisits classical notions such as Krashen's Comprehensible Input Hypothesis in light of contemporary neurocognitive findings, and shows how dual-channel and ecological perspectives further strengthen the conceptual rationale for a two-phase protocol.

2.1 Cognitive Load Theory and the Architecture of Working Memory
Cognitive Load Theory (CLT) provides the primary mechanistic foundation for the STR phase structure. CLT assumes that learning is constrained by the limited capacity of working memory, which can handle only a small number of informational units at any given time. Effective instruction must therefore manage the interplay among:
intrinsic load (the inherent complexity of the material),
extraneous load (load introduced by suboptimal instructional design), and
germane load (the effort invested in building and refining schemas).
Phase I of STR is explicitly designed to **minimize extraneous load**. Interlinear glosses, L1 mediation, dual-channel presentation, and selective textual enhancement reduce the effort required to decode unfamiliar lexical and syntactic material. This frees up cognitive resources for constructing a coherent mental representation of the text.
Phase II reverses the emphasis. By removing scaffolding and requiring learners to reconstruct meaning from memory, it **raises germane load** in a controlled way. This aligns with CLT's view that durable schema construction is driven by effortful, meaning-oriented processing rather than by repeated passive exposure.
The innovation is not the mere presence of support, but the **structured regulation of load across phases** and the explicit treatment of the phase transition as a cognitive-load inflection point. In this sense, STR is not simply a set of helpful techniques; it is a cognitively optimized protocol that times and shapes the learner's workload.

2.2 Sociocultural Theory and the Zone of Proximal Development

Vygotsky's concept of the Zone of Proximal Development (ZPD) offers a complementary rationale for STR's reliance on temporary scaffolding. Within the ZPD, learners can perform tasks with assistance that they cannot yet complete independently; scaffolding functions to bridge this gap and to support the gradual internalization of new skills.

Phase I operationalizes ZPD principles by providing **just-in-time support** that allows learners to understand texts that would otherwise exceed their unaided capabilities. STR extends traditional ZPD applications in two important ways:

Operationalization of scaffolding

STR introduces measurable constructs such as the **L1-Dependency Index** and the **Fading Threshold**, making it possible to empirically test when and how scaffolding supports movement through the ZPD, rather than treating scaffolding as a purely metaphorical concept.

Integration with cognitive load constraints

STR does not assume that more assistance is always better. It emphasizes that support must not generate excessive extraneous load. Scaffolding should be light enough to preserve working memory resources, yet strong enough to enable successful performance—an optimization problem well suited to adaptive, AI-driven systems.

In short, ZPD provides the socio-cognitive rationale for a supported phase (Phase I), while CLT defines the boundary conditions that determine **how much** and **what kind of** support is appropriate at any given moment.

2.3 Revisiting and Reframing Comprehensible Input

Krashen's Comprehensible Input Hypothesis has long framed meaningful input at level *i+1* as the primary engine of language acquisition. While this perspective has shaped reading pedagogy for decades, it has been criticized for conceptual vagueness, limited falsifiability, an overemphasis on passive exposure, and insufficient attention to the mechanisms of memory consolidation.

STR retains the core insight that comprehension-friendly input reduces anxiety and facilitates engagement, but it **reframes** comprehensible input within a broader cognitive architecture. In STR, comprehensible input is treated not as a self-sufficient driver of acquisition, but as a **necessary precondition** for more powerful learning processes that occur during retrieval-based reading in Phase II. Neurocognitive studies indicating that active retrieval and feedback-driven reconstruction engage wider and more robust neural networks than passive comprehension further support this reframing.

Although the Comprehensible Input Hypothesis remains influential, recent theoretical and empirical work has further qualified its sufficiency. Lightbown and Spada (2021) and Long (2015, 2023) argue that while meaning-bearing input at i+1 is necessary for noticing, durable acquisition typically requires interactionally modified input and opportunities for pushed output that trigger deeper processing and negotiation for meaning. Meta-analytic evidence similarly reveals that purely receptive exposure yields significantly smaller long-term lexical gains ($g \approx 0.25$–0.35) than conditions combining comprehension with retrieval or production practice (Webb, Uchihara, & Yanagisawa, 2023; Kim & Webb, 2022). STR therefore positions Phase I comprehensible input as an essential seeding mechanism rather than the primary driver, aligning with contemporary interactionist and usage-based extensions of the original hypothesis.

2.3.1 Contemporary Neurocognitive Critiques

While STR preserves Krashen's central intuition that comprehension-oriented input and reduced anxiety are beneficial, contemporary neurocognitive research challenges the sufficiency of *passive* exposure as a main driver of acquisition. Neuroimaging work on reading and vocabulary learning shows that active, goal-directed language processing—such as production, monitoring, and error-based adjustment—recruits broader fronto-temporo-parietal networks and leads to more robust plastic changes than mere exposure to input (e.g., Braid & Richlan, 2022; Perdue, 2021; Shen, Li, & Sun, 2021). Social interaction and feedback-driven tasks further amplify consolidation, with stronger and more stable activation patterns than those observed under conditions of isolated, silent comprehension.

STR directly addresses these critiques by **reframing comprehensible input as a necessary precondition rather than a self-sufficient mechanism**. Phase I is designed to secure accessibility and stabilize an initial situation model of the text, while Phase II imposes active, effortful retrieval and reconstruction demands. In neurocognitive terms, Phase I provides the low-anxiety, meaning-focused input required to seed representations; Phase II supplies the high-engagement, feedback-sensitive processing needed to strengthen and integrate those representations into long-term memory. Recent syntheses of neuroimaging work on context-based vocabulary learning confirm that effortful retrieval and feedback-driven reconstruction recruit broader fronto-temporo-parietal and hippocampal networks than passive comprehension alone, resulting in more robust and durable neural consolidation (Shen, Li, & Sun, 2021; Braid & Richlan, 2022).

2.4 Dual-Channel Processing and Multimedia Learning Principles
The Cognitive Theory of Multimedia Learning (CTML) adds another layer of justification for the Phase I design. CTML posits that learners process information through at least two partially independent channels—auditory-verbal and visual-pictorial—each with limited capacity. Instruction that distributes information sensibly across these channels can reduce cognitive load and facilitate understanding.
In Phase I, STR leverages CTML by incorporating, where appropriate:
synchronized audio narration with the written text,
visual glosses or icons for key items,
typographical cues (such as color or emphasis) to highlight critical information,
restrained multimodal enhancement that supports, rather than distracts from, reading.
At the same time, STR explicitly acknowledges meta-analytic findings showing that too much enhancement can backfire, creating visual noise and adding extraneous load. Recent second-round meta-analyses confirm that carefully designed multimodal glosses (text + image/audio) outperform text-only glosses on immediate post-tests ($g \approx 0.40$–0.50), yet the long-term advantage largely disappears when retrieval practice is included in both conditions (Mahdi, Mohsen, & Almanea, 2024; Ramezanali et al., 2021). This directly supports STR's minimalist, retrieval-equalizing design.

2.5 Ecological and Neuro-Interactive Perspectives: Reading as Action

Recent ecological and neuro-interactive approaches conceptualize learning as an embodied, action-oriented process in which the environment offers affordances—possibilities for action—that learners can perceive and exploit. From this vantage point, reading is not a passive consumption of text, but a dynamic interplay of perception, attention, and cognitive action.
STR aligns naturally with this view by treating each phase as structuring a distinct **field of affordances**:
Phase I creates an affordance for comprehension by simplifying the perceptual and cognitive landscape and making meaning immediately accessible.
Phase II creates an affordance for autonomous retrieval and reconstruction by reintroducing productive difficulty and requiring the learner to act on previously formed representations.
This ecological lens situates STR within current cognitive science and provides a more realistic account of how learners actually interact with texts: not as passive recipients of input, but as agents navigating and exploiting a designed instructional environment.

2.5.1 STR as Structured Affordance Design

From an **ecological–dynamic** perspective on language learning (Gibson, 1979; van Lier, 2004), STR is not primarily a vehicle for "delivering information," but a way of **structuring fields of affordances** for the learner. In this view, learning emerges from perception–action couplings: the environment offers opportunities for understanding and acting, and the learner selectively actualizes these opportunities based on current abilities and goals.

Within this framing, each STR phase constitutes a distinct affordance configuration:

- **Phase I affordance.** The supported version of the text affords comprehension through in-flow L1 support and glossing. By lowering decoding demands and stabilizing a situation model, it enables successful perception–action coupling (reading, interpreting, anticipating) at the learner's current capacity threshold.
- **Phase II affordance.** The removal of support restructures the environment so that it now affords **autonomous retrieval and reconstruction**. The same text becomes an opportunity to act on previously formed representations: learners must actively recall meanings, monitor understanding, and repair gaps without external linguistic aids.

This ecological framing positions STR as an **action-oriented architecture** rather than a passive input system. It aligns with embodied and neuro-interactive accounts of reading, which emphasize that durable learning arises when learners repeatedly engage in goal-directed, effortful interaction with text, rather than merely receiving information.

Emerging embodied and 4E (embodied, embedded, enacted, extended) perspectives further enrich this view by treating language comprehension as sensorimotor simulation rather than abstract symbol manipulation (Atkinson, 2010; Reggin, Lee, & Vlach, 2023). Phase I scaffolds can therefore usefully incorporate restrained multimodal and sensorimotor cues (e.g., simple images or gestural prompts) that ground meaning in perceptual experience, while Phase II's unsupported reconstruction encourages learners to actively re-enact those simulations from memory. This pattern strengthens form–meaning mappings in line with embodied accounts of L2 lexical consolidation (Macedonia, 2019; Reggin et al., 2023).

2.6 Synthesis: Why a Two-Phase Model is Necessary

Taken together, these theoretical strands converge on three key points:
Comprehension and retention rely on different cognitive conditions and cannot be optimized simultaneously within a single, undifferentiated instructional format.
Scaffolding must be both temporary and efficient, supporting the learner only until comprehension is stabilized and avoiding unnecessary extraneous load.
Retrieval must be deliberately induced if initial understanding is to be transformed into flexible, long-term knowledge.
STR formalizes these insights in the form of a two-phase protocol. It offers the SLA field a **structured, theoretically grounded, and operationally measurable framework** for reading-based instruction that explicitly manages the transition from supported comprehension to autonomous retrieval.
Individual learner differences are likely to moderate the magnitude and trajectory of STR effects. Proficiency, working-memory capacity, and L1–L2 distance influence both the optimal density of Phase I scaffolding and the speed of L1-dependency reduction (Yanagisawa, Webb, & Uchihara, 2020; Indrarathne & Kormos, 2018). Lower-proficiency learners and those with limited working memory appear to benefit most from dense, L1-inclusive support in Phase I followed by strict binary fading, whereas advanced learners may require leaner scaffolding to avoid under-challenge (Jeon & Yamashita, 2014; Suzuki & DeKeyser, 2017). Future implementations—especially AI-driven ones—should therefore incorporate baseline profiling and adaptive thresholds while preserving the core categorical nature of the Phase I → Phase II transition.

Table 1. Empirical foundations of STR by theoretical principle

Principle	Key empirical evidence / meta-analyses (illustrative)	Implication for STR (Phase I / Phase II)
Comprehensible Input & Input Enhancement	Work on comprehensible input and input enhancement shows that meaning-focused, level-appropriate input facilitates initial comprehension and lowers affective barriers, while textual enhancement can increase noticing but with modest effects and possible overload when overused (e.g., Krashen, 1985; Lee & VanPatten, 2003; Han et al., 2008).	**Phase I:** Justifies provision of meaning-oriented, comprehensible text with selective enhancement to support accurate initial understanding without excessive visual noise. **Phase II:** Comprehension achieved in Phase I becomes the baseline for retrieval; input alone is not expected to drive long-term gains without subsequent recall.
Repetition and Retrieval Practice (Testing Effect)	Robust evidence shows that repeated retrieval outperforms repeated study for long-term retention of vocabulary and concepts, especially when retrieval is effortful and spaced (e.g., Roediger & Karpicke, 2006; Karpicke & Blunt, 2011; Pan & Rickard, 2018).	**Phase I:** Acts as the “study” phase that stabilizes an initial representation. **Phase II:** Rereading the same text without support operationalizes retrieval practice in context, turning repetition into a structured testing effect rather than mere re-exposure.
Context-Based / Extensive Reading	Extensive and context-based reading interventions generally show small-to-moderate benefits for vocabulary, fluency, and comprehension, though results are heterogeneous and depend on text difficulty and support (e.g., Elley, 1991; Grabe, 2009; Nakanishi, 2015).	**Phase I:** Embeds scaffolding into continuous, narrative texts, leveraging contextual predictability and story engagement. **Phase II:** Uses the same passages for autonomous rereading, aligning STR with extensive reading while adding a controlled consolidation mechanism.
In-Flow Support (Glossing, Interlinear, Hypertext)	Meta-analyses on glossing and hypertext support indicate that line-proximal, multimodal glosses facilitate incidental vocabulary gains and comprehension, particularly for lower-proficiency learners (e.g., Chun & Plass, 1996; Watanabe, 1997; Yoshii, 2006; Ko, 2012). However, excessive glossing and split attention can increase extraneous load.	**Phase I:** Justifies dense but carefully targeted in-flow support (interlinear glosses, brief L1 cues) to reduce decoding costs and secure comprehension. **Phase II:** Removal of all support tests whether in-flow aids have led to internalized form–meaning mappings or merely transient dependence on external help.
Motivation, Self-Efficacy, and Autonomy	SLA motivation research and self-determination theory show that perceived competence, autonomy, and meaningful progress predict persistence and long-term engagement (e.g., Dörnyei & Ryan, 2015; Ryan & Deci, 2020; Bandura, 1997).	**Phase I:** Provides early, reliable success (“I can understand this text”) and reduces anxiety, supporting competence beliefs. **Phase II:** Creates a structured opportunity for learners to experience unsupported success (“I can do this on my own”), reinforcing autonomy and feeding into an Ideal L2 Self trajectory.
Cognitive Load Management (CLT, Dual-Channel)	CLT and multimedia learning studies show that learning is optimized when extraneous load is minimized and germane load is increased at appropriate times; distributing information across channels can help, but overloading either	**Phase I:** Minimizes extraneous load via well-designed scaffolds and optional dual-channel support (text + audio, simple visuals), ensuring that working memory is devoted to constructing meaning. **Phase II:** Intentionally increases germane load by removing

Principle	Key empirical evidence / meta-analyses (illustrative)	Implication for STR (Phase I / Phase II)
	channel impairs performance (e.g., Sweller, 1988; Mayer, 2005).	aids and inducing desirable difficulty, promoting schema consolidation rather than new extraneous demands.

2.7 Glossing Evidence and Design Implications for STR

Research on glossing and computer-mediated annotations provides important design constraints for STR, particularly for Phase I. Across recent meta-analyses and second-round syntheses (Yanagisawa, Webb, & Uchihara, 2020; Mahdi et al., 2024), several robust patterns emerge. First, **interlinear and line-proximal glosses** tend to outperform marginal or end-of-page glosses, especially for lower-proficiency learners, because they reduce the need for eye movements away from the text and lower split-attention costs. Second, **multimodal glosses** (e.g., L2 form + L1 translation + image or audio) often yield stronger vocabulary gains than single-mode glosses, provided that the additional modes are simple and tightly aligned with the target item. Third, **L1-based glosses** appear particularly beneficial for beginners and lower-intermediate learners, who otherwise expend substantial resources on decoding and inferencing before any deeper processing can occur.

At the same time, the glossing literature also warns against a naïve “more is better” approach. Studies that have manipulated gloss density and complexity report that **overly rich or visually noisy glossing** can increase extraneous cognitive load, slow reading, and in some cases encourage **shallow processing**, where learners skip or only superficially attend to the L2 stream while relying on L1 paraphrases (as noted, for example, in work by Taylor, Boers, and colleagues). These findings support a design stance in which glosses are (a) interlinear and in-flow, (b) multimodal but minimalist, and (c) selectively targeted at genuinely unfamiliar or pedagogically central items rather than applied indiscriminately.

For STR, these results translate into clear design implications. In **Phase I**, glossing should be **in-flow, L1-inclusive, and optionally multimodal**, but carefully throttled to avoid clutter and to keep the learner’s primary focus on the L2 text. In **Phase II**, all glosses are removed, which not only enables retrieval practice but also serves as a diagnostic: if learners cannot reconstruct meaning without the glosses, then Phase I glossing may have functioned as a crutch rather than a bridge. In this way, STR both capitalizes on the strengths of glossing and builds in a structural safeguard against its main risks.

2.7.1 Avoiding Enhancement Overload

While STR employs visual enhancement (interlinear color-coding) to support noticing in Phase I, **meta-analytic evidence on Textual Enhancement (TE)** cautions against excessive layering of visual cues. A recent synthesis of TE studies (e.g., Han, Park, & Combs, 2008) reports that:

- Using **multiple TE techniques simultaneously** (e.g., bold + italics + color) yields an overall effect of around $g \approx 0.54$ (small-to-medium).
- Importantly, **single-technique enhancement was not significantly less effective** than multi-technique enhancement.
- The most plausible interpretation is a **ceiling effect**: once a certain visibility threshold is reached, additional visual signals no longer boost noticing and may instead introduce **extraneous cognitive load** by adding visual noise.

STR design response. In light of these findings, STR deliberately adopts a **minimalist enhancement strategy** in Phase I:

- Uses **single-color, in-flow interlinear glosses** rather than stacked combinations of bolding, underlining, and multiple colors.
- Avoids "decoration overload" (e.g., simultaneous bold, underline, pop-ups for the same item) that would fragment attention and increase split processing.
- Allows enhancement density to be **adaptive to proficiency**: denser support for beginners, sparser and more selective glossing for advanced learners.

STR therefore treats enhancement as a *precise, low-noise signal* rather than a maximalist highlighting strategy.

2.7.2 Enhancement Density and the Ceiling Effect

Meta-analytic evidence on Textual Enhancement (TE) suggests that **"more highlighting" does not necessarily mean "more learning."** A recent synthesis (Han, Park, & Combs, 2008) reports a small–medium overall effect for TE, but **no meaningful advantage** for multiple, stacked techniques over a single, well-chosen cue.

Table 2. Effect sizes for single vs. multiple textual enhancement techniques

TE condition	Typical implementation	Aggregate effect size
Single-technique enhancement	Color only (e.g., L2 forms in one color)	$g \approx 0.48$
Multiple-technique enhancement	Color + bold + italics (and/or underline)	$g \approx 0.54$

Note. Adapted from Lee & Thomson (2025). The difference between conditions was not statistically significant, suggesting a ceiling effect.

Critically, the difference between $g \approx 0.48$ and $g \approx 0.54$ was **not statistically significant**, and moderator analyses pointed to a likely **ceiling effect**: once a form is sufficiently salient, additional visual cues do not increase noticing, but may **inflate extraneous cognitive load** by cluttering the visual field.

STR design decision.
In light of these findings, STR adopts a **minimalist enhancement strategy** in Phase I:

- use **single-mode enhancement** (color-coded, in-flow interlinear glosses);
- avoid stacked cues (simultaneous color + bold + italics + underlining) for the same item;
- reserve additional formatting (e.g., italics) for specific, high-value comments rather than as a global signal.

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3. STR Model Architecture — Phase I, Phase II, and Transitional Dynamics

The Supported Two-Phase Reading (STR) model is not a loose bundle of techniques such as glossing, translation, and rereading. It is a **protocolized instructional sequence** with clearly delineated phases, a discrete transition point, and trackable process indicators. This section specifies the architecture of STR in operational terms, explaining how Phase I (Supported Reading) and Phase II (Consolidation Practice) are sequenced, how scaffolding is *faded*, and how core constructs such as L1-dependency and autonomy are conceptualized.

3.1 STR as a Protocol Rather than a Technique

Within STR, a single reading episode for a given passage is defined as a **four-step chain**:

1. **Phase I – Supported Reading**
   The learner reads the passage with **full in-flow linguistic support**: interlinear translations and brief glosses are embedded directly in the text for all relevant units so that meaning is immediately available.
2. **Categorical Fading Point**
   At the end of Phase I, a **predefined, binary switch** is enacted: all linguistic aids (translations, glosses, color-coding) are removed. For that passage, support shifts from *approximately* 100% to 0%.
3. **Phase II – Retrieval-Based Reading**
   The learner reads the **same passage again**, now completely without linguistic support. Meanings must be reconstructed from memory, thereby engaging retrieval practice in a naturalistic reading context.
4. **Process Recording**
   The episode is documented with respect to comprehension, L1 reliance, cognitive load, and performance, enabling longitudinal tracking within learners and rigorous comparison across conditions.

The protocol is **passage-bound**: each coherent segment (e.g., 200–700 words, depending on proficiency level) is processed through the full P1 → Fade → P2 cycle. STR is thus defined by three essential features:

- comprehensive, in-flow support in Phase I,
- a **categorical removal** of support before Phase II, and
- a **mandatory second reading without aids**.

Conceptually, this four-step chain operationalizes Vygotsky's ZPD: intensive, temporary guidance followed by independent performance; and it implements CLT by first minimizing extraneous load, then intentionally increasing germane load through retrieval.

3.2 Phase I: Supported Reading (Comprehension and Load Reduction)

Phase I is dedicated to securing comprehension while minimizing extraneous cognitive load. It implements the principles of comprehensible input, CLT, and ZPD through dense but cognitively economical scaffolding.

3.2.1 Instructional Goal

The primary objectives of Phase I are to:

- ensure immediate intelligibility of the text,
- prevent early overload and frustration,
- stabilize an initial situation model of the passage, and
- prepare the ground for effective retrieval in Phase II.

Comprehension in Phase I is **not** the endpoint of learning; it is the starting condition for subsequent consolidation.

3.2.2 Interlinear Support and Layout

In STR, support is **interlinear and in-flow**, not marginal or external:

- translations and glosses appear directly after the lexical item or phrase they explain,
- support is visually distinguished (e.g., color, parentheses) but does not require layout switching or page-jumping,
- the reading direction remains linear; split attention is minimized.

This design reflects empirical findings that interlinear and line-proximal glosses reduce eye regressions and processing effort relative to marginal glosses or end-notes, particularly for lower-proficiency learners.

3.2.3 Scope and Type of Support

Phase I typically offers:

- L1 translations for unknown or low-frequency items (especially at A1–A2 levels),
- brief glosses (e.g., base form, core meanings, key valency patterns),
- occasional morphosyntactic cues (e.g., case/aspect labels in morphologically rich languages),
- optional multimodal support (audio, simple images) where appropriate.

Support is **dense but not indiscriminate**. Very high-frequency items, cognates, or transparent forms may be left unglossed to avoid visual clutter and encourage direct L2 processing.

3.2.4 Illustrative Example (Phase I)

The following example illustrates Phase I with German (L1) support for an English (L2) passage:

> When Mr. Hiram B. Otis, the American Minister (als Mr. Hiram B. Otis, der amerikanische Gesandte), bought Canterville Chase, everyone told him (Schloss Canterville kaufte, warnten ihn alle; *to buy, bought, bought – kaufen; chase – das Jagdschloss, das Landgut; hier: Schloss Canterville*) it was a bad idea (es wäre eine schlechte Idee). The place had a ghost (dort soll es gespukt haben). There was no doubt about that (daran bestand kein Zweifel; *doubt – der Zweifel*). Lord Canterville was a man (Lord Canterville war ein Mann) who paid attention to detail (der großen Wert auf Details legte; *to pay attention to smth. – Aufmerksamkeit schenken, aufpassen, achten auf, sich kümmern um; to pay – bezahlen; attention – die Aufmerksamkeit, die Achtung, die Beachtung*). He felt it was his duty (er sah es als seine Pflicht an; *to feel, felt, felt – fühlen, sich fühlen, sich befinden, meinen, spüren, glauben*) to tell Mr. Otis about the ghost (Mr. Otis über den Geist zu informieren; *to tell, told, told – sagen, erzählen, berichten*). This happened (dies ereignete sich; *to happen – passieren, geschehen*) when they sat down (als sie sich setzten; *to sit down – sich setzen; to sit, sat, sat – sitzen*) to discuss the terms (um über die Kaufbedingungen zu sprechen; *to discuss – diskutieren, besprechen, erörtern; terms – Bedingungen, Konditionen*).

Color-coding and in-flow placement allow learners to consult translations instantly, or to ignore them when meaning is already known. Phase I thus merges **intensive support** with **extensive reading principles**, as passages form part of a longer narrative that increases exposure density, reduces extraneous load through contextual predictability, and supports motivation via story engagement.

3.3 Phase II: Consolidation Practice (Retrieval-Based Reading)

Phase II shifts the instructional focus from comprehension to **retrieval, consolidation, and autonomy**. It operationalizes the testing effect within a reading-based format.

3.3.1 Instructional Goal

The main goals of Phase II are to:

- require learners to retrieve meanings and constructions without external linguistic aids,
- strengthen memory traces through effortful recall ("desirable difficulty"),
- reduce long-term dependence on L1 or other support languages, and
- provide observable indicators of emerging autonomy.

Where Phase I asks, *"Can the learner understand with help?"*, Phase II asks, *"Can the learner reconstruct meaning without help?"*

3.3.2 Procedure and Example

Immediately following Phase I, the same passage is presented again:

- all translations, glosses, and color-coding are removed;
- the visual layout is **identical**, apart from the absence of aids;
- learners read for meaning, relying on what was processed in Phase I.

For the example above, Phase II presents the passage as:

When Mr. Hiram B. Otis, the American Minister, bought Canterville Chase, everyone told him it was a bad idea. The place had a ghost. There was no doubt about that. Lord Canterville was a man who paid attention to detail. He felt it was his duty to tell Mr. Otis about the ghost. This happened when they sat down to discuss the terms.

This second, unsupported pass implements **retrieval practice in context**. Additional optional tasks may accompany Phase II, such as:

- brief comprehension questions,
- L2 paraphrases of key sentences,
- short written summaries,
- targeted retrieval of specific lexical items.

However, the **minimal definition** of Phase II within STR is a second, unsupported pass over the identical text under conditions of 0% external linguistic support.

3.3.3 Flexible Application

For beginners, the canonical order is Phase I (with support) → Phase II (without support). More advanced learners may invert this sequence—first attempting Phase II unaided, then consulting Phase I as a reference. This flexibility allows STR to be aligned with different proficiency levels without changing the underlying architecture of *supported comprehension followed by aid-free retrieval*.

3.4 Fading Scaffolding in STR: Binary, Passage-Bound Transition

A central innovation of STR is the formalization of *fading scaffolding* as a **categorical, auditable event**, rather than a vague, continuous process.

3.4.1 Definition

In the specific sense used in STR, *fading scaffolding* denotes a **phase-based, categorical removal of all linguistic aids** (translations and glosses) between an initial supported reading (Phase I) and a subsequent reading (Phase II) of the *same* passage. Its key properties are:

- **Binary:** support is approximately 100% in Phase I and 0% in Phase II for that passage; there are no intermediate support levels within the same episode.
- **Passage-bound:** the fading event is tied to a specific text segment and does not depend on moment-to-moment learner choice.
- **Pre-specified:** the fading moment is determined in advance and can be preregistered in empirical studies.

This sharply distinguishes STR from continuous or adaptive fading systems, where support density is gradually reduced but rarely reaches an explicitly enforced 0%.

3.4.2 Rationale for Binary Fading

The binary fading mechanism serves three purposes:

1. **Theoretical clarity**
   It allows precise testing of the hypothesis that complete removal of support between two passes over the same passage yields superior delayed retention and faster decline in L1 dependence.
2. **Methodological auditability**
   Because 0% support in Phase II is unambiguous, adherence can be verified via **layout audits** (no gloss regions in Phase II), eye-tracking (no fixations on gloss areas in Phase II), and system logs (no access to Phase I during Phase II).
3. **Pedagogical salience**
   For learners, the fading moment becomes a visible and meaningful transition: support has ended; autonomy is now expected. Successful completion of Phase II can thus enhance self-efficacy and perceived competence.

3.4.3 Fixed vs Adaptive Fading

In the base protocol, fading occurs after exactly one fully supported pass (Phase I) over the passage. This keeps time-on-task and exposure constant across conditions in experimental comparisons. In applied or AI-supported implementations, fading may become **adaptive**, informed by:

- Phase I comprehension indicators,
- cognitive load metrics (e.g., pupillary response, reading time),
- error rates on intra-phase checks,
- learner self-reported confidence.

In all cases, what remains invariant is the **categorical nature** of the transition: once Phase II begins, support for that passage is 0%.

3.5 Core Process Constructs: L1-Dependency, Autonomy, and Fading Threshold

The STR architecture introduces several constructs that capture the process of moving from supported to autonomous reading.

3.5.1 L1-Dependency

L1-dependency refers to the extent to which a learner relies on their first language (or another support language) during or immediately after reading. Conceptually, STR treats L1-dependency as a **passage-level, ordinal construct** that can be captured, for example, on a 0–3 scale:

- **0 – No observable L1 recourse**
  Phase II tasks are completed entirely in L2, without consulting L1 or producing L1 paraphrases.
- **1 – Brief, incidental L1 use**
  Short, occasional L1 checks that do not dominate processing.
- **2 – Extended L1 reliance**
  Frequent or prolonged L1 lookups or L1-based reconstruction of meaning.
- **3 – Task dependence on L1**
  Phase II tasks cannot be completed without systematic recourse to L1 support.

In empirical work, this categorical index can be supplemented by continuous indicators (time spent on aids, language of responses, eye-tracking metrics) and modeled as a latent trajectory across multiple STR episodes.

3.5.2 Autonomy

Autonomy captures how well learners function without support, **controlling for** what they already understood with support. Conceptually, autonomy can be defined as:

Phase II performance (e.g., comprehension, lexical recall, paraphrasing quality), **adjusted for** Phase I comprehension and baseline L2 proficiency.

This framing emphasizes that autonomy is not raw ability, but **the capacity to maintain performance in an unsupported phase once comprehension has been assured in the supported phase**.

3.5.3 Fading Threshold

The *fading threshold* represents the minimal conditions under which it is pedagogically and cognitively reasonable to withdraw support. These conditions may be specified in terms of:

- minimum Phase I comprehension (e.g., a threshold on comprehension checks),
- acceptable cognitive load (no signs of catastrophic overload),
- stabilization or reduction of L1-dependency within Phase I.

In fixed protocols, the fading threshold is standardized (e.g., always after one complete supported pass). In adaptive systems, it may be operationalized as a combination of performance and load indicators, potentially informed by AI-based analytics.

3.6 Granularity and Text Segmentation

Finally, the STR architecture assumes a specific **granularity of application**:

- texts are segmented into coherent passages (e.g., one paragraph, a short scene, or 200–700 words),
- each passage is processed through the full P1 → Fade → P2 cycle,
- for beginners (A1–A2), shorter passages with denser support are recommended; for intermediate learners (B1–B2), longer passages with more selective glossing may be used.

Across a full book or course, **hundreds of such micro-cycles** can be chained, creating repeated opportunities for supported comprehension followed by autonomous retrieval. In this way, STR operationalizes a **micro-genetic view** of reading development: each passage-level cycle becomes a small experimental unit in which support, fading, and retrieval can be observed, measured, and optimized over time.

4. Empirical and Technological Extensions of STR — AI Integration, Measurement, and Research Agenda

The STR framework is currently positioned as a **conceptually well-specified but empirically untested protocol**. Its long-term relevance depends on two parallel developments: (a) the accumulation of rigorous empirical evidence and (b) the creation of scalable technological implementations that embody the protocol in real learning environments. This section outlines how AI can serve as both a *research instrument* and a *pedagogical engine* for STR, and proposes a research agenda that translates the conceptual model into falsifiable hypotheses, measurable constructs, and concrete study designs.

4.1 AI-Supported Content Generation and Personalization

One of the primary practical bottlenecks for STR is the **generation of high-quality, two-phase reading materials**: phase-aligned passages, accurate interlinear glosses, and carefully controlled layout. Large Language Models (LLMs) and related AI tools can substantially reduce this burden.

4.1.1 Automated STR Material Generation

AI systems can be configured to:

simplify authentic texts to a target proficiency level,

generate interlinear translations and brief glosses for Phase I,

output a corresponding Phase II version of the passage without any aids,

maintain strict P1–P2 identity at the surface level (identical wording, punctuation, and segmentation).

Such pipelines can be implemented as an **STR Annotation System**, where texts are processed into dual-phase HTML or ebook formats with consistent color-coding, structural markers, and metadata for research (e.g., target words, part-of-speech tags, frequency bands).

Human oversight remains essential: automatic translations and glosses require editorial review for accuracy, idiomaticity, and appropriateness. However, AI drastically reduces the *first-pass* workload, enabling large-scale creation of STR-conformant materials across multiple languages and domains.

4.1.2 Personalized Text Selection and Thematic Relevance

Beyond static materials, AI can dynamically tailor STR content to individual interests and goals:

learners can submit any text (news, fiction, professional documents),

the system simplifies and annotates it into a dual-phase format,

vocabulary can be selected to align with learner-specific lexical profiles.

This **interest-driven personalization** is particularly important from a motivational perspective: when learners engage with topics they care about, the two-phase cycle (support → retrieval) is more likely to be sustained over time.

4.2 AI-Enhanced Scaffolding and Adaptive Fading

The base STR protocol specifies a fixed sequence: one fully supported pass (Phase I) followed by one unsupported pass (Phase II) over the same passage. AI integration does not alter this fundamental architecture, but it enables adaptive control over (a) how much support is shown in Phase I, (b) **when** fading is triggered, and (c) **how** feedback is delivered in Phase II. In other words, AI allows STR to move from a static protocol to a data-driven, learner-sensitive implementation while preserving its core two-phase logic.

4.2.1 Dynamic Glossing and Learner Modeling

In the baseline STR format, Phase I provides fixed-density support: all preselected target items are glossed for all learners. AI-based learner modeling can refine this by tailoring glossing to individual profiles. A simple learner model can:

- **Estimate lexical status.**
  Track responses, lookup behavior, and exposure history across sessions to estimate for each item a probability of being unknown or fragile, *p*(unknown).
- **Display glosses selectively.**
  Show glosses only for items with *p*(unknown) above a threshold (e.g., 0.7), suppressing glosses for items that are likely known. This reduces visual clutter and potential overload while preserving support where it is most needed.
- **Adapt glossing across cycles.**
  Gradually fade glosses for items that show stable retrieval in Phase II across multiple passages, effectively "weaning" learners off L1 support for those forms.

The result is an individualized support profile: beginners receive relatively dense scaffolding, whereas more advanced learners encounter a leaner, more L2-dominant text—even within the same STR framework and the same curricular materials.

4.2.2 Adaptive Fading Thresholds

AI-based analytics can also inform **when** to trigger the fading moment, i.e., the transition from Phase I to Phase II. Rather than imposing a strictly fixed threshold (after one supported pass for all learners and texts), an adaptive STR system can adjust the fading point on the basis of multiple indicators, such as:

- **Phase I comprehension performance**
  Brief local checks or embedded questions indicating whether the core meaning of the passage has been established.
- **Behavioral markers**
  Reading time, regressions, and gloss access patterns that signal either smooth processing or ongoing struggle.
- **Self-reported confidence**
  Quick ratings at passage or session level that complement behavioral data.
- **Physiological data (advanced setups)**
  Task-evoked pupillary response or similar measures as proxies for cognitive load.

If these indicators suggest that comprehension is unstable or cognitive load is already high, the system can delay Phase II, temporarily reduce text difficulty, or introduce an intermediate micro-support step (e.g., partial fading for only a subset of items) **before** the fully unsupported pass. Conversely, if Phase I performance is strong and load moderate, the system can trigger earlier fading or increase the difficulty of subsequent passages (e.g., fewer glosses, syntactically denser input).

Example implementation.

- A learner in Phase II shows **low accuracy** (e.g., < 60% on comprehension/recall items) and **high latency** (e.g., > 3 seconds per target item).
- The system flags this passage as **"above current ZPD band"** and marks the fading point as premature.
- In the next session, the AI adapts by either
  - auto-simplifying the syntax of the upcoming passage at the same storyline point, **or**

    - inserting an **intermediate transitional step**: a brief additional pass with *partial* glossing on the most problematic items, followed by a proper, fully unsupported Phase II pass.

Crucially, these adaptive fading thresholds preserve the **categorical nature** of STR's architecture: once Phase II has begun for a given passage, support remains at 0%. AI merely helps decide when that categorical switch is pedagogically optimal for a particular learner and text, and whether an extra semi-supported micro-step is needed *before* that switch.

4.2.3 Intelligent Feedback in Phase II

During Phase II, AI can act as an intelligent tutoring layer that monitors retrieval performance and provides targeted support **without** reintroducing full L1 scaffolding. Possible functions include:

- **Misunderstanding detection.**
  Analyzing responses to comprehension questions, paraphrases, or cloze items to detect systematic gaps or misinterpretations.
- **Targeted feedback and examples.**
  Providing concise explanations or additional in-context examples for fragile items (e.g., collocations, idioms, polysemous verbs) when errors or hesitations occur.
- **Micro-drills and adaptive spacing.**
  Generating brief drills (e.g., sentence completions, minimal pairs) for problematic items and scheduling their reappearance across future passages according to spacing principles.
- **Metacognitive prompts.**
  Encouraging learners to reflect on which words still require L1 support and which can already be handled autonomously, reinforcing the perception of progress.

Phase II feedback loop (illustrative scenario).

- A learner repeatedly confuses the pair **"affect"** vs **"effect"** in Phase II tasks.
- The AI system detects this error pattern and triggers a micro-intervention consisting of:
    - a brief context comparison (e.g., *"The medicine affected her sleep"* vs. *"The effect was immediate"*),
    - a 3-item retrieval quiz focusing specifically on this contrast, and
    - scheduled **spaced re-exposure** of the pair in subsequent passages, with light prompts until performance stabilizes.

In this configuration, STR becomes a **closed feedback loop**: initial support in Phase I → retrieval in Phase II → AI-based diagnosis → targeted reinforcement in later cycles. AI does not replace the two-phase logic of STR; it **refines** the deployment of support and challenge so that fading and feedback are systematically tuned to the learner's evolving state.

4.3 Measurement Infrastructure: From Constructs to Data

4.3.1 Core Outcome Measures

At minimum, STR-oriented research should track the following outcome domains:
Vocabulary gains
Immediate and delayed recall/recognition of target items
Distinction between form recall, meaning recall, and receptive recognition

Reading comprehension
Global (text-level) and local (sentence-/proposition-level) comprehension measures
Performance differences between Phase I and Phase II

Cognitive load
Subjective ratings (e.g., brief NASA–TLX variants)
Single-item task difficulty rating per passage (1–7 Likert)
Behavioral indicators (e.g., reading time per 100 words, regression counts)
Optionally, physiological measures (pupillometry, eye-tracking)
Cognitive load will be assessed using a **multi-method approach**, combining subjective and behavioral indicators. In addition to a brief NASA–TLX variant, participants will provide a single-item task difficulty rating (1–7 Likert) for each focal passage. These ratings will be triangulated with behavioral measures such as reading time per 100 words (and, where available, eye-tracking or pupillometry), allowing researchers to distinguish efficient low-load comprehension in Phase I from **"desirable difficulty"** in Phase II.

L1-dependency and autonomy
Frequency and duration of L1 use
Language choice in oral and written responses
Performance in Phase II statistically adjusted for Phase I comprehension and baseline L2 proficiency

4.3.2 L1-Dependency Index and Validation

The **L1-Dependency Index** can be operationalized as a composite construct based on:
system logs (time spent viewing glosses, use of translation buttons),
eye-tracking (fixations on gloss regions in experimental settings),
qualitative indicators (L1 notes, L1-based paraphrases).
To validate this index, it should be triangulated with established instruments such as:
LEAP-Q (Language Experience and Proficiency Questionnaire),
BLP (Bilingual Language Profile),
BSWQ (Bilingual Switching Questionnaire).
These tools serve not merely as covariates but as **anchors** for convergent and discriminant validity, ensuring that the L1-Dependency Index captures a genuine dimension of bilingual processing rather than generic proficiency or motivation.

4.3.3 Autonomy Score and Longitudinal Tracking
The **Autonomy Score** should be modeled as a longitudinal construct that combines:
Phase II performance (across tasks and sessions),
reduced L1 reliance over time,
stability or improvement in comprehension under decreasing support.
Mixed-effects modeling and item-response theory can be used to:
control for prior knowledge and text difficulty,
estimate growth trajectories,
compare STR with alternative conditions (e.g., interlinear-only, repeated reading, marginal glosses).
AI-based platforms can compute these metrics automatically and present them both as **research data** and as **learner-facing dashboards**, illustrating progress in a motivating and interpretable way.

4.3.4 Operationalization of the L1-Dependency Index and Autonomy Score

While the previous subsections have outlined L1-dependency and autonomy as core process constructs, their usefulness depends on being operationalized in ways that are both **practical** and **psychometrically defensible**. This subsection proposes a provisional measurement strategy to be refined in future empirical work.

L1-Dependency Index (0–3 Episode-Level Scale)

At the episode level (i.e., for a given passage and its associated Phase I → Phase II cycle), L1-dependency can be captured with a simple 0–3 rating scale synthesizing behavioral indicators of L1 use:

- **0 – No observable L1 recourse**
  The learner completes Phase II (unsupported reading and associated tasks) entirely in the L2, without consulting translations, writing L1 notes, or giving L1-based responses.
- **1 – Minimal, incidental L1 use**
  Occasional, brief L1 checks (e.g., a quick dictionary glance or an isolated L1 note) that do not substantially affect task performance or time-on-task.
- **2 – Extended but non-essential L1 reliance**
  Regular or prolonged L1 use (e.g., frequent lookups, longer L1 notes, partial L1 paraphrasing) that clearly supports performance, but where the learner can still complete tasks largely in the L2.
- **3 – Task-level dependence on L1**
  The learner is unable to complete Phase II tasks without systematic recourse to L1 support (e.g., full-sentence L1 translations, continuous dictionary use, predominantly L1 responses).

In digital implementations, this ordinal index can be complemented by continuous indicators automatically logged by the system (time spent in gloss views, number of translation clicks, language of typed responses) and, in experimental settings, by eye-tracking measures (fixations on support regions). Together, these sources provide a rich basis for modeling L1-dependency as a **latent construct** across multiple episodes.

MTMM Validation Strategy (Expanded)

The L1-Dependency Index requires **construct validation** that goes beyond face validity. We therefore propose embedding it in a **multitrait–multimethod (MTMM)** design.

Traits (what is being measured)

1. **L1-dependency during reading (STR-specific)** – episode-level reliance on L1 within the STR protocol.
2. **General L1 reliance in bilingual contexts (established construct)** – habitual dominance, switching, and reliance on L1 across everyday situations.

Methods (how it is being measured)

1. **Behavioral** – time on glosses, translation clicks, and eye-tracking metrics (fixations, regressions) derived from STR system logs.
2. **Self-report** – standardized questionnaires such as **LEAP-Q**, **BLP**, and **BSWQ** capturing language history, dominance, and switching patterns.
3. **Performance-based** – language choice and accuracy in Phase II tasks (e.g., L2 vs. L1 responses, quality of L2 paraphrases).

Validation logic

- **Convergent validity.**
  The L1-Dependency Index should correlate **moderately** ($r \approx 0.40$–0.60) with indicators of L1 dominance and switching behavior derived from LEAP-Q, BLP, and BSWQ (e.g., LEAP-Q L1 dominance scores, BSWQ switching frequency in L1-dominant contexts).

- **Discriminant validity.**
  The L1-Dependency Index should correlate only **weakly** ($r < 0.30$) with general cognitive ability (e.g., working memory span) and should **diverge** from measures of L2 proficiency (e.g., CEFR level, standardized vocabulary tests). In other words, it should not reduce to "low proficiency" or "low ability".
- **Predictive validity.**
  Baseline L1-Dependency (e.g., Week 1) should meaningfully predict Phase II performance at later time points (e.g., Week 8), with regression coefficients of at least $\beta \geq 0.40$, **controlling for baseline proficiency**. Higher initial L1-dependency is expected to predict slower growth in autonomy unless STR is effective in reducing it.

Role of LEAP-Q, BLP, and BSWQ.
Crucially, LEAP-Q, BLP, and BSWQ are not treated merely as secondary covariates. They function as **primary anchors for convergent and discriminant validity**, establishing that the L1-Dependency Index taps a genuine dimension of bilingual processing (context-specific L1 reliance) rather than generic proficiency, motivation, or exposure.

Autonomy Score as Adjusted Phase II Performance

The Autonomy Score is intended to quantify how well learners function without support, while controlling for their initial comprehension and overall proficiency. Conceptually, it can be defined as:

Phase II performance statistically adjusted for Phase I comprehension and baseline L2 proficiency.

In practice, this can be implemented in several complementary ways:

- **Episode level.**
  Autonomy can be modeled as the **residual** of Phase II performance (e.g., comprehension accuracy, lexical recall, paraphrase quality) after regressing out Phase I performance on the same passage and standardized proficiency scores.
- **Longitudinal level.**
  Mixed-effects or latent growth models can estimate individual intercepts and slopes for Phase II performance over time, including Phase I performance, proficiency, and motivational covariates as predictors. The resulting random effects can be interpreted as autonomy parameters.

By construction, an autonomy measure defined in this way reflects how much **additional unsupported performance** a learner demonstrates beyond what would be expected from their initial supported comprehension and general L2 level. This makes it possible to test one of STR's central claims: that repeated exposure to the Phase I → Phase II cycle should, over time, yield **measurable gains in autonomy**, even when overall proficiency is held constant.

Synthesis

Taken together, (a) a carefully specified L1-Dependency Index, (b) an MTMM-based validation strategy using LEAP-Q/BLP/BSWQ as **primary instruments**, and (c) an Autonomy Score defined as adjusted Phase II performance provide a concrete methodological foundation for treating "reduced L1 reliance" and "increased autonomy" not as vague pedagogical ideals, but as **quantifiable outcomes** in a systematic program of STR research.

4.4 Research Agenda: From Hypothesis to Falsification

The STR framework yields a set of **explicit, preregistrable hypotheses** that can anchor a multi-stage research program. Together, they target process, outcomes, moderators, mechanisms, and affective consequences.

4.4.1 L1-Dependency Trajectory (Hypothesis 1)

Core claim.
The categorical removal of support between Phase I and Phase II accelerates the decline in L1 reliance compared with structurally similar approaches that never enforce a fully support-free phase.

Hypothesis 1 (L1-dependency slope).
Over an 8-week intervention, learners in STR conditions will show a **significantly steeper negative slope** on the L1-Dependency Index than learners in interlinear-only, marginal-gloss, or assisted-repeated-reading conditions, under **matched exposure time and text difficulty**.

Falsification.
If STR does **not** produce a steeper decline in L1-dependency than comparison conditions (or produces a flatter or reversed slope), the central claim that binary fading uniquely reduces L1 reliance is not supported.

4.4.2 Temporal Specificity of STR Effects (Hypothesis 2)

Core claim.
Re-reading the same passage without support immediately after fully supported reading functions primarily as a consolidation mechanism. Consequently, STR is expected to yield **temporally asymmetric** benefits: negligible at the immediate post-test, but increasingly positive at longer delays.

Hypothesis 2 (retention profile).
Under equal exposure time and matched materials, the expected retention profile is:

- **Immediate post-test (T0, same day).**
  No meaningful difference is expected between STR and an interlinear-only condition (STR ≈ interlinear-only). At this point, both formats provide strong in-flow support, and initial encoding is primarily driven by Phase-I–like scaffolding in both groups.
- **Short delay (T1, 3–4 days).**
  A **small but emerging** advantage is anticipated for STR, with effect sizes of approximately $d \approx 0.20$–0.30 (STR > interlinear-only), particularly for vocabulary and form–meaning recall. This pattern reflects early consolidation benefits of Phase II retrieval while memory traces are still relatively fresh.
- **Long delay (T2, 4 weeks).**
  A **clearly non-trivial advantage** is expected for STR, with effect sizes at least in the small-to-moderate range (SESOI: $d \geq 0.30$), and plausibly around $d \approx 0.40$–0.50, especially on delayed vocabulary and phrase recall (STR > interlinear-only). At this stage, retention is presumed to depend strongly on prior retrieval, and the absence of a mandatory retrieval phase in interlinear-only conditions should manifest in accelerated forgetting.

Proposed mechanism.
Under this hypothesis, Phase II functions primarily as a **retrieval-based consolidation phase**, not as an additional study episode. Phase I (in both STR and interlinear-only conditions) stabilizes an initial situation model of the text under high support and drives immediate performance at T0. In STR, the Phase II rereading without aids requires learners to reconstruct meaning from memory, strengthening lexical and constructional representations. As a result, group differences are predicted to emerge in the **shape of the retention trajectory** (T0 → T1 → T2), rather than in immediate post-test scores.

Falsification criterion.
Hypothesis 2 will be considered **not supported** if, at the long-term delay (T2, 4 weeks), STR fails to show at least a small-to-moderate advantage over the comparison condition. Concretely, if the between-group effect size on delayed vocabulary outcomes is $d < 0.30$ and the 95% confidence interval includes zero, the retrieval-based consolidation mechanism claimed for STR's long-term benefit is not corroborated.

4.4.3 Moderator Analysis: Gloss Modality Effects (Hypothesis 2.1)

Background.
Meta-analyses (e.g., Ramezanali et al., 2021; Mahdi et al., 2024) suggest that **multimodal glosses** (text + image + audio) often outperform text-only glosses on **immediate** tests, whereas long-term advantages are smaller or inconsistent. STR provides a structured context to test this as a **moderator** of Hypothesis 2.

Hypothesis 2.1 (Modality × Retention Interval interaction).

- At **T0 (immediate)**, STR with multimodal glosses in Phase I will yield **higher** vocabulary scores than STR with text-only glosses ($\approx d$ 0.40).
- At **T2 (4 weeks)**, the difference between multimodal and text-only STR conditions will be **negligible** ($\approx d$ 0.10, 95% CI including zero).

Rationale.
Multimodal glosses enhance **encoding** in Phase I; however, Phase II requires **active retrieval** in both conditions, which should **equalize long-term retention**. If Phase II functions as intended, text-only glosses may be **sufficient** for durable learning, making STR more cost-effective to implement.

Falsification.
If multimodal STR retains a clear advantage at long delay (e.g., $d \geq 0.30$ at 4 weeks), the claim that Phase II retrieval largely compensates for simpler Phase I support must be revised, and multimodal design may be necessary for maximal effect.

4.4.4 Cognitive Load Profiles (Hypothesis 3)

Core claim.
STR optimizes **cognitive load distribution**: it reduces extraneous load in Phase I and increases germane load in Phase II to a **"desirable difficulty"** range, rather than to overload.

Hypothesis 3 (load redistribution).
Relative to repeated reading without structured support, STR will show:

- **Phase I:** Lower extraneous load indicators (shorter fixations and fewer regressions on key segments, lower NASA–TLX scores, lower task difficulty ratings), reflecting efficient supported comprehension.
- **Phase II:** A **moderate increase** in effort markers (higher task difficulty ratings, somewhat longer reading times, modestly higher NASA–TLX scores), and these increases will correlate **positively** with delayed retention, indicating desirable, not destructive, difficulty.

Falsification.
If STR does not reduce extraneous load in Phase I, or if Phase II effort markers are either flat (no desirable difficulty) or associated with **worse** learning (indicative of overload), the load-optimization rationale of STR is challenged.

4.4.5 Motivation and Persistence (Hypothesis 4)

Core claim.
The two-phase cycle—**assured comprehension** in Phase I followed by **visible autonomy** in Phase II—should enhance learners' sense of competence and autonomy, leading to higher motivation and persistence.

Hypothesis 4 (motivational and behavioral outcomes).
Compared with traditional extensive reading or non-protocolized glossed reading, STR groups will:

- show **lower attrition** across multi-week interventions;
- report **larger gains** on competence/autonomy–related subscales of motivational instruments (e.g., self-efficacy, perceived reading competence, autonomy);
- demonstrate **greater behavioral persistence**, such as higher uptake of optional Phase-II–like tasks or continued use of STR materials beyond mandatory requirements.

Falsification.
If STR does not outperform control conditions on persistence and competence/autonomy measures—or if it leads to **higher dropout** due to perceived difficulty—the motivational advantage attributed to its two-phase architecture is not supported.

Taken together, these hypotheses can be **preregistered** with explicit SESOI thresholds and analysis plans. This positions STR not merely as a conceptual framework, but as a **genuinely falsifiable model**, inviting systematic confirmation or disconfirmation rather than post-hoc justification.

4.5 Dual-Track Development: Scientific Validation and Product Design
To fully realize the potential of STR, empirical research and technological development should proceed in a **dual-track manner**:
Track 1 – Scientific Validation
– design and implement pilot studies (e.g., small-sample lab experiments with eye-tracking),
– scale up to classroom-based randomized controlled trials,
– publish results in peer-reviewed SLA and educational technology journals.
Track 2 – Product-Oriented Implementation (MVP)
– develop a minimally viable STR platform that delivers two-phase reading experiences,
– embed logging and measurement capabilities by design,
– use platform data to refine learner models, fading strategies, and glossing density.

The two tracks are mutually reinforcing: the platform generates ecologically valid data for research, while research findings inform iterative improvements in the platform's design. Scientific credibility becomes not just an academic asset but a **central value proposition** in a crowded EdTech landscape.

In summary, AI technologies and modern measurement tools do not replace the theoretical core of STR; they **instantiate and extend it**. They make it possible to generate materials at scale, personalize scaffolding, precisely control fading, and continuously track learning trajectories. The next step is to integrate these components into concrete study designs and implementations, which subsequent sections address through comparative analyses, limitations, and future directions.

4.6 Falsifiability and Open Science Implementation

For STR to function as a genuine scientific model rather than a collection of attractive intuitions, its core claims must be formulated in ways that allow them to be **clearly supported or clearly not supported** by data. This requires (a) specifying **a priori criteria for falsification** and (b) embedding these criteria into **preregistered, open-science research designs**.

We propose at least four families of falsifiable predictions and corresponding decision rules:

1. **L1-Dependency Trajectory**
   A central process claim of STR is that the categorical removal of support between Phase I and Phase II accelerates the decline in L1 reliance, relative to structurally similar conditions that do not enforce an aid-free rereading. Formally, STR predicts a **steeper negative slope** on the L1-Dependency Index over time for STR learners than for learners in interlinear-only, parallel-text, or repeated-reading conditions under matched exposure and difficulty.
   - *Falsification criterion*: If preregistered longitudinal analyses (e.g., mixed-effects models) show no reliably more negative slope for STR, or if the difference in slopes remains below a pre-specified smallest effect size of interest (SESOI), STR's process-level claim about L1-dependency reduction is not supported.
2. **Delayed Retention and SESOI for Learning Outcomes**
   At the outcome level, STR predicts superior delayed retention (e.g., 3–4 weeks) of target vocabulary and constructions compared to study-only or non-structured rereading conditions, given equal time-on-task and text difficulty.
   - *Falsification criterion*: Prior to data collection, a SESOI (e.g., **d = 0.30** for delayed vocabulary or constructional retention) is specified. If the STR advantage on delayed post-tests is statistically indistinguishable from zero *and* the observed effect size falls below this SESOI (with compatible confidence intervals), the claim that STR confers practically meaningful retention benefits is not supported.
3. **Protocol Integrity: 0% Aids in Phase II**
   The definitional claim of STR is that Phase II constitutes reading under **0% external linguistic support** for the target passage. This is not a theoretical nuance but a protocol requirement that must be enforced and verified.
   - *Integrity criterion*: Preregistration should specify manipulation checks such as layout audits, log-based verification that no glosses or translations were available in Phase II, and, where possible, eye-tracking confirmation that no support regions were present or fixated. Studies that fail these checks cannot be treated as valid tests of STR; evidence from such studies is considered **uninformative** with respect to STR's core claims.
4. **Classroom Feasibility and Attrition Thresholds**
   STR is intended for real-world instruction, not only for laboratory settings. It therefore makes a practical claim: that the two-phase protocol can be implemented without unacceptable attrition or non-compliance.
   - *Feasibility criterion*: Preregistrations should define tolerable bounds for dropout and protocol non-adherence (e.g., maximum acceptable attrition rate, proportion of missed

Phase II readings). If implementation in typical classroom settings systematically exceeds these thresholds, STR may be judged **pedagogically impractical**, even if its effects are theoretically promising.

All of these criteria—and their associated SESOIs, analysis plans, and manipulation checks—should be **specified in advance** via preregistration (e.g., on OSF or a comparable platform). Preregistrations should include:

- explicit hypotheses and alternative models,
- outcome and process measures (with primary vs. secondary distinctions),
- SESOIs and planned contrasts,
- sample size justifications and stopping rules,
- protocol-adherence checks and exclusion criteria,
- a commitment to report null and disconfirming results.

Finally, STR research should adhere to open science practices wherever ethical and feasible: sharing **materials** (annotated STR texts), **analysis code**, and **de-identified data** to support reanalysis, replication, and meta-analytic synthesis. In this way, the model is not only conceptually falsifiable, but also embedded in a research culture where falsification is practically possible—and where strong, converging evidence in favor of STR would carry genuine weight.

4.7 Cognitive Load Measurement and Triangulation

In empirical implementations of STR, cognitive load will be assessed using a **multi-method package** that combines subjective ratings, behavioral indicators, and, where feasible, physiological data. The aim is to move beyond reliance on a single questionnaire and to triangulate the construct of mental effort in reading-based tasks.

First, **subjective mental effort** will be measured with a brief variant of the NASA-TLX, administered at the passage or session level to capture global mental effort associated with STR tasks. Second, **task-specific difficulty** will be indexed via a single-item rating for each focal passage on a 1–7 Likert scale (e.g., "How difficult was this passage to process?"). This rating complements NASA-TLX by targeting perceived challenge at the level of the individual reading episode rather than the entire session.

Third, **behavioral indicators** such as reading time per 100 words (and, in laboratory settings, eye-movement metrics like regressions) will be used as objective indices of processing difficulty. Where available, these may be supplemented by **physiological indicators**, in particular task-evoked pupillary response, as an additional proxy for working-memory load.

This triangulation strategy is intended to enhance the validity of cognitive load measurement in STR studies. NASA-TLX provides a global estimate of mental effort, the task difficulty rating captures passage-specific challenge, and behavioral measures offer external validation of these subjective reports. The expected pattern is that Phase I will be characterized by relatively low perceived difficulty and low effort scores (efficient supported comprehension), whereas Phase II will exhibit moderately higher difficulty ratings (approximately 4–5 on a 7-point scale) and increased effort—consistent with the notion of **desirable difficulty** rather than cognitive overload.

4.8 AI-Based STR Annotation Pipeline

To make STR practically deployable at scale, the model requires an efficient way to convert ordinary texts into Phase I / Phase II materials with consistent formatting and high linguistic quality. An AI-based **STR Annotation Pipeline** can automate most of this work. In its simplest form, such a pipeline takes an input text (e.g., a chapter from a graded reader or an authentic article), segments it into passage-sized units, and then uses a large language model to generate Phase I versions with in-flow glosses, brief L1 translations, and optional multimodal cues, alongside Phase II versions that preserve the exact same wording, punctuation, and segmentation but contain **no** linguistic support. Automated quality checks—such as HTML validation, strict P1–P2 identity checks at the token level, and consistency checks for gloss formatting—are used to ensure that the output complies with the STR protocol. Human review can then focus on spot-checking semantic accuracy and pedagogical appropriateness rather than manually annotating every passage.

In line with open-science principles, the full technical specification of this STR Annotation System (e.g., prompts, pipeline stages, quality-control criteria, and a checklist for annotation quality) is best documented separately, for example in an **online technical report or OSF repository**, with illustrative code and sample corpora made publicly available. In the present article, the pipeline is therefore treated not as a closed commercial tool, but as a **reproducible reference implementation** that can be adapted, audited, and improved by other researchers and practitioners who wish to generate STR-conformant materials in different languages and instructional settings.

5. Positioning STR Among Existing Reading Approaches

The Supported Two-Phase Reading (STR) protocol is part of a broader ecosystem of input- and reading-based approaches, including extensive reading, glossing, interlinear and parallel texts, and repeated reading. This section situates STR within that landscape and highlights where it converges with, and diverges from, its closest relatives—structurally (how support is organized), functionally (what the learner is asked to do), and methodologically (how the approach can be tested).

5.1 Comparative Overview of STR and Related Approaches

Table 2 summarizes key similarities and differences between STR and several widely used reading approaches. The focus is on support configuration, the presence or absence of a mandated retrieval phase, and the extent to which each approach can be treated as a standardized, researchable protocol.

Table 3. STR in comparison with existing reading-based approaches

Approach	Typical support configuration	Fading / withdrawal of support	Retrieval status	Layout & cognitive load features	Relation to STR
Extensive Reading (ER)	Light or no support; graded texts, meaning-focused input	No systematic fading; support usually minimal from the outset	Retrieval largely incidental; rereading not required	Low visual load; minimal interruption; high variability in text–learner match	STR adopts ER-style long-form, narrative texts but adds dense Phase I support and a mandatory, unsupported Phase II rereading.
Glossing (marginal / hypertext)	Local glosses in margins, footnotes, or pop-ups; often	Support remains available throughout reading; learner	Retrieval not structurally enforced; may occur if learners	Split attention if glosses are distant from the text; hypertext may	Phase I of STR uses in-flow, interlinear glossing to reduce extraneous load; Phase II removes all glosses,

Approach	Typical support configuration	Fading / withdrawal of support	Retrieval status	Layout & cognitive load features	Relation to STR
	L1 or L2 explanations	decides whether to consult	reread or avoid glosses	reduce but not eliminate this	turning what is usually permanent support into temporary scaffolding.
Interlinear Reading (interlinear-only)	Dense, line-by-line L1 equivalents directly under L2; strong guarantee of comprehension	No built-in fading; learners may never read without the interlinear layer	Retrieval optional; many learners process mainly via L1	In-flow support, low split attention, but high risk of chronic L1 dependence	STR retains the in-flow advantages of interlinear formats in Phase I but adds binary, protocolized fading and a mandatory L2-only Phase II, with L1-dependency monitored over time.
Parallel Texts / Bilingual Editions	L2 and L1 versions in separate columns or pages; full-sentence translations	No fading; L1 text remains permanently available	Retrieval optional; learners can repeatedly check L1	Frequent layout switching; potential alignment difficulties between L2 and L1 segments	STR avoids parallel layout by embedding support inside the L2 stream (Phase I) and then removing it entirely (Phase II), reducing split attention and enforcing L2-only processing.
Repeated Reading (RR) / Assisted RR	Same text read multiple times; sometimes with audio modeling or teacher support	No standard rule for when or whether support is reduced	Retrieval may occur but is not structurally specified as the main objective	Cognitive load depends on text choice and teacher implementation; often focused on fluency	STR resembles RR in reading the same passage twice, but Phase I guarantees comprehension and Phase II is explicitly defined as unsupported retrieval, not just another reading.
Popular interlinear methods (e.g. Ilya Frank)	L2 text with embedded L1 segments and later “original” text without glosses	Second, unsupported reading is recommended but typically optional; degree of support varies	Retrieval possible but neither mandatory nor monitored	In-flow support, often verbose L1 paraphrases; low initial difficulty but risk of shallow L2 processing	STR can be seen as an academically specified, testable reformulation: Phase II is obligatory, fading is precisely defined (100% → 0%), and outcomes (L1-dependency, autonomy) are operationalized.
STR (Supported Two-Phase Reading)	Phase I: dense, in-flow support (interlinear glosses, L1 cues, optional	Binary, passage-bound fading between Phase I and Phase II;	Retrieval is the central aim of Phase II; aid-free	Phase I minimizes extraneous load while ensuring comprehension;	STR integrates elements of all above approaches into a unified, protocolized

Approach	Typical support configuration	Fading / withdrawal of support	Retrieval status	Layout & cognitive load features	Relation to STR
	multimodal input). Phase II: same passage with 0% support	timing can be fixed or adaptive but is always explicit	rereading is mandatory	Phase II increases germane load via desirable difficulty	cycle with process measures and a clear research agenda.

5.2 Approach-Specific Notes

Several points of comparison are particularly salient:

- **Extensive Reading vs STR**
  STR aligns with ER in using long-form, narrative texts and prioritizing meaning-focused engagement. It departs from ER by adding **dense scaffolding in Phase I** and a **mandatory retrieval-based rereading in Phase II**, turning what is often a purely input-oriented practice into a comprehension–consolidation cycle.
- **Glossing / Interlinear vs STR**
  STR inherits the strengths of **in-flow, interlinear glossing**—high comprehension and low split attention—while addressing the main risk: chronic L1 dependence. Under STR, such support is confined to Phase I and is **guaranteed to disappear** at the fading point. The subsequent Phase II then serves as both consolidation and a test of whether glosses promoted genuine learning or merely temporary reliance on L1.
- **Parallel Texts vs STR**
  Parallel texts are best viewed as enduring reference tools: the L1 version remains available and is not structurally tied to any retrieval demand. STR, in contrast, embeds support directly in the L2 stream and then **removes it altogether**, reducing layout-induced extraneous load and forcing the learner to reconstruct meaning within the L2 channel.
- **Repeated Reading vs STR**
  Both RR and STR involve multiple passes over the same passage, but RR typically does not guarantee initial comprehension nor define a clear support-withdrawal rule. STR can therefore be characterized as **"repeated reading with controlled scaffolding and enforced retrieval"**: Phase I ensures that there is something meaningful to retrieve; Phase II defines retrieval as the instructional target rather than a by-product.
- **Popular interlinear methods vs STR**
  Methods such as the Ilya Frank approach combine "reading with support" and "reading without support" at the level of user experience, but lack a standardized definition of how much support is provided, when it is removed, or how success is measured. STR reframes this intuition as a **research protocol** with a mandatory Phase II, a clearly defined fading event, and **explicit process constructs** (L1-Dependency Index, Autonomy Score) that can be tracked across time.

5.3 Fair-Comparison Logic for Future Studies

Positioning STR among existing approaches is not only conceptual, but also methodological. To evaluate whether STR offers genuine advantages over ER, interlinear-only reading, parallel texts, or RR, comparative studies must adhere to a **fair-comparison logic**, including:

- **Time- and exposure parity**
  Competing conditions should be matched on total reading time, number of encounters with target items, and overall text difficulty. STR should not be given more time-on-task or qualitatively

easier texts than comparison conditions if the goal is to attribute differences in outcomes to the protocol itself.

- **Protocol integrity and layout audits**
  For interlinear and parallel-text comparisons, it is essential to verify that:
  (a) STR's Phase II truly occurs at 0% support (no glosses or translations available), and
  (b) alternative conditions adhere to their own design (e.g., interlinear-only support remains present throughout). Layout audits and, where feasible, eye-tracking can serve as manipulation checks, ensuring that differences in outcomes are not artefacts of unplanned exposure or hidden support.
- **Outcome and process symmetry**
  Analyses should consider both **outcome measures** (e.g., delayed vocabulary retention, comprehension) and **process measures** (e.g., L1-Dependency Index, Autonomy Score), so that STR is not judged solely on immediate score gains but also on its ability to reduce L1 reliance and foster autonomous reading.

5.4 Summary: STR as a Bridging, Protocolized Framework

Across these comparisons, three aspects of STR are distinctive:

1. **Integrated dual function** – It unites comprehension assurance (Phase I) and retrieval-based consolidation (Phase II) within a single, standardized cycle.
2. **Binary, auditable fading** – It treats the removal of support as a discrete, measurable event rather than a gradual, informal trend.
3. **Process-oriented measurement** – It introduces constructs and indices (e.g., L1-Dependency, Autonomy) that make the transition from supported to autonomous reading empirically traceable.

Rather than replacing extensive reading, glossing, interlinear or repeated reading, STR **reorganizes and couples their strengths** into a coherent, testable architecture that can be evaluated under fair, time- and exposure-matched conditions.

6. Limitations, Open Questions, and Conclusion
No conceptual model is without constraints. While STR offers a theoretically rich and methodologically precise framework, its current formulation faces several limitations and open questions. Recognizing these boundaries is essential for guiding both empirical work and practical implementation.

6.1 Conceptual and Pedagogical Limitations

6.1.1 Focus on Receptive Skills

STR primarily targets **reading comprehension and receptive vocabulary**. Any impact on productive skills (speaking, writing) is indirect, mediated through improved lexical access and form–meaning mappings. Without complementary output-oriented activities (e.g., retelling, summarizing, discussion), gains in oral and written proficiency may be limited.

6.1.2 Dependence on Translation and Risk of L1 Overreliance

Although Phase II is designed to counteract long-term L1 dependence, Phase I's heavy use of L1 support carries inherent risks:
learners may prefer to "skim" L2 and read glosses instead,
shallow processing may occur if attention remains on L1,
some learners may resist transitioning to Phase II, especially at low proficiency or high anxiety levels.
The protocol mitigates these risks by enforcing an aid-free phase and monitoring L1 usage, but **overuse or misalignment of Phase I** (e.g., excessive duration, inappropriate text difficulty) can still foster a persistent L1 crutch.

6.1.3 Sensitivity to Text Difficulty and Selection

STR's effectiveness depends strongly on **text–learner match**:

texts that are too difficult may overload learners even in Phase I,
texts that are too easy reduce the need for scaffolding and weaken the rationale for STR,
genres with highly fragmented or dialogic structure may be less suitable for clear passage-level cycles.
Careful text curation and level calibration are therefore crucial, especially in classroom implementations.

6.1.4 Limited Cultural and Pragmatic Scope

Reading-based approaches, including STR, are only partially suited to transmitting **pragmatic norms, interactional routines, and embodied aspects of communication**. Without additional tasks (e.g., role-plays, multimodal input, real-world interaction), learners may develop strong receptive skills but comparatively weaker sociopragmatic competence.

6.2 Empirical and Methodological Limitations

6.2.1 The Empirical Gap and Pathways to Validation

The primary limitation of STR in its current form is what may be termed an **empirical vacuum**: although the protocol is theoretically well grounded and operationally specified, **all effectiveness claims are, at present, hypothetical**.

To date, no controlled studies have tested whether:

- **Binary fading** outperforms **gradual fading**;
- the **L1-Dependency Index** declines faster under STR, or
- **Phase II** retrieval yields $d \geq 0.30$ at delayed tests.

This situation should not be interpreted as a flaw of the model, but rather as a **developmental stage**. STR is introduced in this article as a **conceptual and positioning contribution** that establishes:

1. **A falsifiable protocol** with clearly preregistrable hypotheses about fading, retrieval, and L1-dependency;
2. **Operational definitions** for key constructs (e.g., L1-dependency, autonomy, fading moment) that can be measured with existing instruments and logging tools;
3. **A roadmap for systematic empirical validation**, specifying experimental contrasts, outcome measures, and SESOI benchmarks.

The research program outlined in Chapter 7 is explicitly designed to move STR from a **"promising framework"** to an **"evidence-based practice"** through staged validation, beginning with proof-of-concept pilots and progressing to larger, preregistered, and independently replicable studies.

6.2.2 Measurement and Construct Validity

The proposed indices (L1-dependency, autonomy, fading threshold) are conceptually plausible but methodologically complex:
they require multimethod data (behavior, self-report, eye-tracking, logs),
they must be disentangled from general proficiency, motivation, and working memory effects,
measurement invariance across proficiency levels and language pairs is not yet established.
A substantial portion of the research agenda must therefore be devoted to **construct validation**, not only outcome comparison.

6.2.3 Ecological Validity and Classroom Constraints

Implementing STR in real classrooms raises practical issues:
teachers may have limited time to manage two-phase cycles,
adherence to Phase II (no "cheating" with aids) may be difficult to enforce without digital control,
cluster-randomized trials in school settings face attrition, scheduling, and compliance challenges.

There is a risk that STR will work well in **laboratory conditions** but prove difficult to apply in daily teaching without simplification or adaptation.

6.3 Technological and Ethical Considerations

6.3.1 Dependence on AI and Devices

Many of the envisioned strengths of STR—adaptive fading, dynamic glossing, biometric monitoring—assume access to:
reliable digital devices,
robust AI models,
stable connectivity,
potentially eye-tracking or other sensors.
This raises concerns about **equity and accessibility**: not all learners or institutions will have equal access to such infrastructure.

6.3.2 Data Privacy and Ethical Use

AI-enhanced STR implementations necessarily involve extensive data collection:
reading logs,
error profiles,
potentially biometric signals.
This calls for stringent **data protection, transparency, and informed consent**. Ethical guidelines must govern what data are collected, how they are used, and who controls them.

6.4 Future Directions

Given these limitations, several lines of future work are particularly promising:
Comparative trials: randomized studies comparing STR with interlinear-only reading, repeated reading, ER, and parallel texts under time- and exposure-controlled conditions.
Longitudinal designs: tracking L1-dependency and autonomy across months, not just weeks, to test whether Phase II effects accumulate.
Construct validation: multimethod studies to validate L1-Dependency and Autonomy as distinct, reliable constructs, including their relation to bilingual experience profiles.
AI prototypes and MVPs: development of STR-based platforms that allow large-scale, ecologically valid data collection while providing learners with immediate benefits.
Extensions to productive skills: integration of Phase II with brief, structured output activities (oral retellings, micro-writing, dialogic tasks) to bridge receptive and productive competence.

6.5 Conclusion

Supported Two-Phase Reading (STR) offers a **coherent, protocolized response** to a long-standing challenge in second language reading: how to combine secure comprehension with durable, autonomous retention. By explicitly separating a supported comprehension phase from an aid-free retrieval phase, and by treating the transition between them as a central mechanism rather than a side effect, STR synthesizes insights from Cognitive Load Theory, sociocultural scaffolding, retrieval practice, and contemporary AI-assisted pedagogy.
At its core, STR is not a "new method" in the sense of introducing entirely novel techniques. Its innovation lies in **how familiar techniques are combined, timed, and measured**: full interlinear support in Phase I, categorical fading, mandatory rereading in Phase II, and process-oriented indices such as L1-dependency and autonomy. This protocolization makes long-standing pedagogical intuitions empirically testable and technologically implementable.
Future research will determine whether STR delivers on its promise: fostering faster reduction in L1 reliance, stronger delayed vocabulary retention, and more robust autonomous reading skills than existing approaches. Regardless of specific effect sizes, STR already contributes a valuable conceptual tool to the SLA field: a **sharply defined, falsifiable framework** for thinking about how comprehension support and retrieval practice can be integrated into a single, cognitively grounded reading architecture.

7. Dual-Track Development Strategy

7.1 The Validation–Implementation Paradox

STR faces a familiar but often under-articulated challenge in educational innovation: **rigorous empirical validation requires implementation infrastructure**, yet **investment in infrastructure typically presupposes preliminary evidence and academic credibility**.

If development proceeds **only** on the scientific track, STR risks remaining a purely conceptual contribution with limited real-world impact. If development proceeds **only** on the product track, STR risks becoming "just another app" without a differentiated, evidence-based value proposition.

To address this **validation–implementation paradox**, we propose a **dual-track development strategy** in which scientific validation and product development advance in parallel and are explicitly designed to be mutually reinforcing.

7.2 Track 1: Scientific Validation Engine

Track 1 positions STR as an empirically testable framework and builds the evidential base required for academic and institutional uptake. A plausible staged plan would include:

- **Phase 1 (Year 1): Pilot studies (N ≈ 40–120)**
    - Small-scale experiments using manually prepared STR materials (print or simple digital formats).
    - Focus on feasibility, manipulation checks (0% support in Phase II), and initial estimates of effect sizes for L1-dependency trajectories and delayed retention.
- **Phase 2 (Year 2): Preregistered RCTs (N ≈ 300–400)**
    - Cluster- or individually randomized trials in classroom or blended-learning environments.
    - Comparisons with interlinear-only, marginal-gloss, and repeated-reading conditions under time- and exposure-matched designs.
    - Predefined SESOI thresholds and falsification criteria (e.g., $d \geq 0.30$ at 4 weeks for delayed retention).
- **Phase 3 (Year 3): Replication and Moderator Analyses**
    - Replication across languages, proficiency levels, and institutional contexts.
    - Focus on moderators such as gloss modality, textual enhancement density, and learner profiles (e.g., L1 dominance, working memory capacity).

Output. Peer-reviewed publications, preregistered reports, and open datasets establishing STR as a **research-based protocol** rather than a purely theoretical suggestion.

7.3 Track 2: MVP Development Engine

Track 2 develops STR as a **usable digital platform**, both to support learning in authentic settings and to provide the infrastructure for Track 1 research. The development can likewise proceed in staged form:

- **Phase 1 (Year 1): Basic digital STR platform**
    - Fixed P1 → P2 sequence, with static interlinear glosses and simple logging (time-on-page, progression, basic comprehension checks).
    - Focus on usability, robustness, and faithful implementation of the core protocol (100% support in Phase I, 0% in Phase II).
- **Phase 2 (Year 2): AI augmentation**

    - Integration of **dynamic glossing** and elementary learner modeling (e.g., selective gloss display based on item difficulty and prior performance).
    - Dashboards for learners and researchers summarizing L1-dependency indicators, autonomy scores, and retention metrics.
- **Phase 3 (Year 3): Adaptive fading and neuro-adaptive features**
    - Implementation of **adaptive fading thresholds** informed by comprehension indicators, behavioral markers, and—where feasible—physiological data (e.g., pupillometry).
    - Advanced analytics for longitudinal modeling of autonomy and motivation.

Output. A research-ready platform that doubles as an **early commercial prototype**, capable of being deployed in institutional settings and integrated into existing LMS ecosystems.

7.4 Synergy Between Tracks and Risks of Single-Track Development

The two tracks are designed to be **mutually dependent and mutually beneficial**:

- **Data flow from Track 2 to Track 1.**
  The MVP/platform continuously generates high-resolution process data (e.g., gloss usage, Phase I/II performance, L1-dependency indicators), which can be fed directly into preregistered studies without the overhead of manual data collection.
- **Theory-to-design flow from Track 1 to Track 2.**
  Findings from pilot studies and RCTs (e.g., on optimal gloss density, modality effects, cognitive load profiles) inform feature prioritization and design decisions in the platform, ensuring that product evolution remains aligned with evidence rather than ad hoc user requests.
- **Credibility as a shared asset.**
  Scientific credibility from Track 1 (peer-reviewed evidence, open materials, preregistered protocols) becomes a **core marketing and adoption asset** for Track 2, distinguishing STR-based tools from purely commercial products. Conversely, wide deployment of the platform strengthens the external validity and ecological relevance of Track 1 findings.

Critically, these tracks should **not** be viewed as sequential (first research, then product) but as **parallel and interlocking**. Attempting commercialization without a robust evidence base risks diluting STR into yet another unvalidated method. Pursuing purely academic research without scalable implementation risks confining STR to theoretical debates.

The dual-track strategy therefore positions STR as a **long-term translational project**: a protocol that is at once **falsifiable in the lab** and **actionable in practice**, with iterative feedback loops between science and product development.

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Appendix B: STR Implementation Checklist

This checklist is intended to help researchers and practitioners implement Supported Two-Phase Reading (STR) in a way that is auditable, reproducible, and suitable for preregistered studies. Items are organized by phase and by data/analysis requirements.

A. Materials and Design

1. **Text selection**
  - ☐ Narrative or expository continuous text (not isolated sentences)
  - ☐ Length appropriate for level (A1--A2: adapted; B1--B2: lightly adapted/authentic)
  - ☐ Target vocabulary identified in advance (frequency, difficulty)
  - ☐ Same text is used in Phase 1 and Phase 2 (word-for-word identity)

2. **Support density (Phase 1)**
  - ☐ Interlinear or in-flow glossing (no margin switching)
  - ☐ Support available for 100% of preselected target items
  - ☐ L1 glosses/translations are accurate and aligned with line/segment
  - ☐ Visual differentiation of support (color, style, or positioning)

3. **Fading definition**
  - ☐ Categorical (binary) removal of all aids in Phase 2
  - ☐ Fading is tied to the *same* passage (not a later chapter)
  - ☐ Learners cannot navigate back to Phase 1 during Phase 2

B. Procedure (Session Level)

4. **Phase 1: supported reading**
  - ☐ Learner reads full passage with integrated support
  - ☐ Minimum time-on-page or embedded comprehension checks to ensure engagement
  - ☐ (Optional) audio support logged separately

5. **Transition / manipulation check**
  - ☐ System or teacher confirms completion of Phase 1
  - ☐ Layout or platform switches to aid-free version of the *same* passage
  - ☐ Check that support density in Phase 2 = 0% (automated or manual)

6. **Phase 2: unsupported rereading**

- ☐ Passage shown without any L1 aids, glosses, pop-ups, or hyperlinks
- ☐ Retrieval required (e.g. comprehension questions, prompted recall, cloze on target items)
- ☐ External help (dictionary, return to Phase 1) is technically blocked or logged

C. Fidelity and Logging

7. **Adherence logging**
   - ☐ Time stamps for Phase 1 and Phase 2
   - ☐ Navigation logs (no backtracking to supported version)
   - ☐ Completion of Phase-2 tasks recorded

8. **Layout / format audit**
   - ☐ Random sample of sessions checked for 0% support in Phase 2
   - ☐ Any residual aids flagged and excluded or analyzed separately

D. Measurement Package

9. **Core outcomes (minimum set)**
   - ☐ L1-Dependency Index (behavioral coding, self-report, and/or gaze data)
   - ☐ Immediate comprehension/recall for the passage
   - ☐ Delayed recall (e.g. 1--4 weeks) on the same target items

10. **Recommended covariates**
    - ☐ Baseline vocabulary / proficiency (e.g. VST)
    - ☐ Reading speed / working memory (for ANCOVA or LMM)
    - ☐ Motivation / autonomy scales (to detect engagement effects)

11. **Protocol integrity indicator**
    - ☐ Binary variable "fading achieved: yes/no"
    - ☐ Sessions with "no" reported separately and, if preregistered, excluded from per-protocol analyses

E. Preregistration and Sharing

12. **Preregistration elements**
    - ☐ Primary outcome specified (e.g. slope of L1-Dependency Index across weeks)
    - ☐ SESOI defined (e.g. *d* = 0.30 for delayed recall)
    - ☐ Analysis model specified (e.g. mixed-effects with random intercepts for participant/item/class)
    - ☐ Handling of missing data described (e.g. MI under MAR; ITT + PP)

13. **Open materials**
    - ☐ Phase-1 and Phase-2 HTML/PDF versions uploaded
    - ☐ Coding manuals for L1-Dependency Index shared
    - ☐ Scripts for automated layout checks (0% support in Phase 2) shared